

Targeted SAT Math — Set 01

Student: Karina

Print this PDF. Work on paper. Calculator / Desmos is fine.

Do not print the last page for her (answer key).

This set is only the leaks from Test 5 and desk **B14**, plus five last-four Module 2–style stretch items. It is not another full mixed packet. Items are instructor-written SAT-style questions, not official College Board stems, and **not** taken from Practice Tests 1–11.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

1. Notes for this set only

Successive percent

- Each change **multiplies**. Do not add the percents.
- Up $p\%$ means $\times \left(1 + \frac{p}{100}\right)$. Down $p\%$ means $\times \left(1 - \frac{p}{100}\right)$.
- Up 10% then up 10% is $\times 1.1 \times 1.1 = 1.21$, a **21%** increase, not $+20\%$.
- Up 10% then down 10% is $\times 1.1 \times 0.9 = 0.99$. You do **not** return to the start.

“Increase by $p\%$ ”

- Increase x by $p\%$ means $x\left(1 + \frac{p}{100}\right)$.
- Increase by 100% = double ($\times 2$). Increase by 400% = $\times 5$.
- “400% **of** x ” is a different sentence ($\times 4$).

Exponential period

- $A(1 + r)^{kt}$ completes k growth cycles in one unit of t .
- If t is in years, months per cycle = $12/k$.
- Example: exponent $2t \Rightarrow 2$ cycles per year $\Rightarrow$ every 6 months.

Tangent to a circle

- The radius to the contact point P is **perpendicular** to the tangent at P .

- Perpendicular slopes: $m_1 m_2 = -1$ (negative reciprocal). Flip the fraction and change the sign.
- Vertical radius $\Rightarrow$ horizontal tangent (slope 0).
- Write the tangent with point-slope **at** P , not at the center. Then test points in that line.

Last-four Module 2 style (items 21–25)

On the paper practice tests, Module 2 #24–#27 are the usual score-splitters. On the real Bluebook exam there are 22 questions per module, so the same role is the last few items. Those questions almost always **stack two skills**. Items 21–25 below are written in that style. They are originals, not pulled from Tests 1–11.

2. Questions

1. (SPR) A club has 400 members. Membership increases 20% in year 1 and increases 20% again in year 2. How many members are there after 2 years?

Answer: __

2. A \$50 price increases by 20%, then the new price decreases by 20%. What is the final price?

- A) \$50
- B) \$48
- C) \$52
- D) \$40

3. The result of increasing the quantity x by 300% is 80. What is the value of x ?

- A) $\frac{80}{3}$
- B) 20
- C) 240
- D) 320

4. (SPR) A population starts at 80 and grows by 25% each year. What is the population after 2 years?

Answer: __

5. The function Q is defined by $Q(t) = 500(1.06)^{2t}$, where t is the number of years after 2020. According to the model, Q increases by 6% every n months. What is the value of n ?

- A) 2
- B) 6

C) 12

D) 24

6. The function P is defined by $P(t) = 300(1.08)^{3t}$, where t is years after 2010. The model increases the population by 8% every n months. What is n ?

A) 3

B) 4

C) 8

D) 36

7. (SPR) A circle has center $(0, 0)$. A line is tangent to the circle at $(3, 4)$. What is the slope of the tangent line?

Answer: __

8. A circle in the xy -plane has center $(2, 3)$. Line ℓ is tangent to the circle at $(6, 6)$. Which point also lies on ℓ ?

A) $(2, 3)$

B) $(3, 10)$

C) $(8, 9)$

D) $(5, 3)$

9. (SPR) A circle has center $(-2, 1)$. A tangent line touches the circle at $(1, 5)$. The equation of the tangent line can be written as $y = mx + b$. What is the value of b ?

Answer: __

10. A radius of a circle runs from $(1, 2)$ to the point of tangency $(5, 4)$. What is the slope of the tangent line at $(5, 4)$?

A) $\frac{1}{2}$

B) 2

C) -2

D) $-\frac{1}{2}$

11. A circle has center $(4, 1)$. A line is tangent to the circle at $(4, 7)$. What is the slope of the tangent line?

A) 0

B) 1

C) undefined

D) 6

12. A jacket costs \\$40. The store marks it up 10%, then takes 10% off the **new** price. How much less than \\$40 is the final price?

- A) \\$0
- B) \\$0.40
- C) \\$4
- D) \\$8

13. (SPR) A balance of \\$200 increases by 10% at the end of year 1 and increases by 10% again at the end of year 2. What is the balance, in dollars, after 2 years?

Answer: ____

14. The result of increasing the quantity x by 150% is 75. What is the value of x ?

- A) 30
- B) 50
- C) 112.5
- D) 187.5

15. A population of 500 grows by 20%, then decreases by 20%. What is the new population?

- A) 500
- B) 480
- C) 520
- D) 400

16. The function S is defined by $S(t) = 90(1.02)^{t/4}$, where t is years after 2018. According to the model, S increases by 2% every n months. What is the value of n ?

- A) 4
- B) 8
- C) 24
- D) 48

17. A circle has center $(3, 0)$. A line is tangent to the circle at $(3, 5)$. What is the slope of the tangent line?

- A) 0
- B) 1
- C) undefined
- D) 5

18. (SPR) A circle has center $(-4, -1)$. A line is tangent to the circle at $(0, 2)$. What is the slope of the tangent line?

Answer: ____

19. A price of \\$120 decreases by 25%, then the new price increases by 25%. What is the final price?
- A) \\$120
 - B) \\$112.50
 - C) \\$115
 - D) \\$90

20. Which value is equivalent to two successive 15% increases?

- A) a 30% increase
- B) $\times 1.30$
- C) $\times 1.3225$
- D) $\times 1.15$

21. (SPR) A circle has center $(1, -2)$. A line is tangent to the circle at $(5, 1)$. The tangent intersects the y -axis at $(0, b)$. What is the value of b ?

Answer: __

22. The function R is defined by $R(t) = 80(1.05)^{4t/3}$, where t is years after 2015. According to the model, R increases by 5% every n months. What is n ?

- A) 4
- B) 8
- C) 9
- D) 16

23. (SPR) A quantity is increased by 50%, then that result is increased by 50% again. The final value is 225. What was the original quantity?

Answer: __

24. Which quadratic equation has no real solutions?

- A) $x^2 - 6x + 9 = 0$
- B) $x^2 - 6x + 5 = 0$
- C) $2x^2 - 3x - 2 = 0$
- D) $3x^2 - 2x + 5 = 0$

25. (SPR) The legs of a right triangle have lengths x and $x + 7$. The hypotenuse is the diameter of a circle. The area of the triangle is 60. What is the diameter of the circle?

Answer: __

3. Answer key (instructor)

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#	Answer	Why
1	576	$400 \times 1.2 \times 1.2 = 576$. Adding 20% twice is the trap (560).
2	B, \$48	$50 \times 1.2 \times 0.8 = 48$. Same percent up and down does not cancel.
3	B, 20	Increase by 300% means $4x = 80$.
4	125	$80 \times (1.25)^2 = 125$. Same move as desk B14.
5	B, 6	Exponent $2t = 2$ cycles per year = every 6 months.
6	B, 4	Exponent $3t = 3$ cycles per year = every 4 months.
7	$-\frac{3}{4}$	Radius slope $\frac{4}{3}$; tangent is the negative reciprocal.
8	B, (3, 10)	Radius slope $\frac{3}{4}$, tangent slope $-\frac{4}{3}$. Line: $y - 6 = -\frac{4}{3}(x - 6)$. Choice A is the center.
9	$\frac{23}{4}$ or 5.75	Radius slope $\frac{4}{3}$, tangent slope $-\frac{3}{4}$. $y - 5 = -\frac{3}{4}(x - 1)$.
10	C, -2	Radius slope $\frac{1}{2}$, so $m_t = -2$. Product is -1.
11	A, 0	Radius is vertical, so the tangent is horizontal.
12	B, \$0.40	$40 \times 1.1 \times 0.9 = 39.60$. Short of \$40 by 0.40.
13	242	$200 \times 1.1 \times 1.1 = 242$.
14	A, 30	Increase by 150% means $2.5x = 75$. Trap B is “150% of 75” thinking inverted.
15	B, 480	$500 \times 1.2 \times 0.8 = 480$.
16	D, 48	Exponent $t/4 =$ one cycle every 4 years = 48 months.
17	A, 0	Radius is vertical, so the tangent is horizontal.
18	$-\frac{4}{3}$	Radius slope $\frac{3}{4}$; tangent is the negative reciprocal.
19	B, \$112.50	$120 \times 0.75 \times 1.25 = 112.5$.
20	C, $\times 1.3225$	$1.15^2 = 1.3225$, not +30%.
21	$\frac{23}{3}$	Radius slope $\frac{3}{4}$, tangent slope $-\frac{4}{3}$. $y - 1 = -\frac{4}{3}(x - 5)$. At $x = 0$, $y = \frac{23}{3}$.

#	Answer	Why
22	C, 9	Exponent $\frac{4}{3}t = \frac{4}{3}$ cycles per year = one cycle every $\frac{3}{4}$ year = 9 months.
23	100	$x \times 1.5 \times 1.5 = 225 \Rightarrow x = 100$.
24	D	Discriminant of D is $4 - 60 < 0$. A is a repeated root; B and C have two real roots.
25	17	$\frac{1}{2}x(x + 7) = 60 \Rightarrow x = 8$. Legs 8 and 15, hypotenuse 17.